

R02: R Basics

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We will open RStudio on our computer (not R!)

1.1.2 Using R via RStudio

Recall our car analogy from earlier. Much as we don't drive a car by interacting directly with the engine but rather by interacting with elements on the car's dashboard, we won't be using R directly but rather we will use RStudio's interface. After you install R and RStudio on your computer, you'll have two new *programs* (also called *applications*) you can open. We'll always work in RStudio and not in the R application. Figure 1.2 shows what icon you should be clicking on your computer.

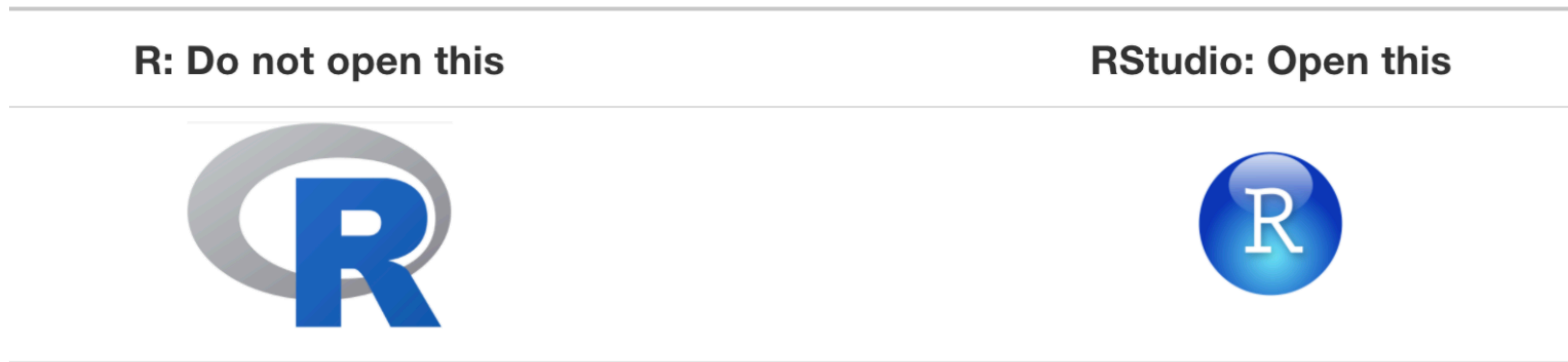


FIGURE 1.2: Icons of R versus RStudio on your computer.

RStudio anatomy

BuzzR

RStudio anatomy

<https://buzzrbeeline.blog/>

Emma Rand

Script file

Write code here
To run code put your cursor on the line and click the **run button**
Edit to correct errors
⇒ record of commands that worked
Save scripts with the **.R** extension
⇒ syntax will be highlighted
⇒ good practice
<- is the assignment operator
⇒ puts what is on the right in to the object on the left
⇒ Assign results if you want to use them again

Console

When you click run, code is sent to the console and executed
> is the prompt
⇒ do not type it
⇒ appears when R is ready for next command
Command output goes here by default
⇒ output is in a different colour
⇒ [1] indicates 3.4 is the first element of the output
⇒ many commands will not have output, the prompt just reappears

Script: where you write code

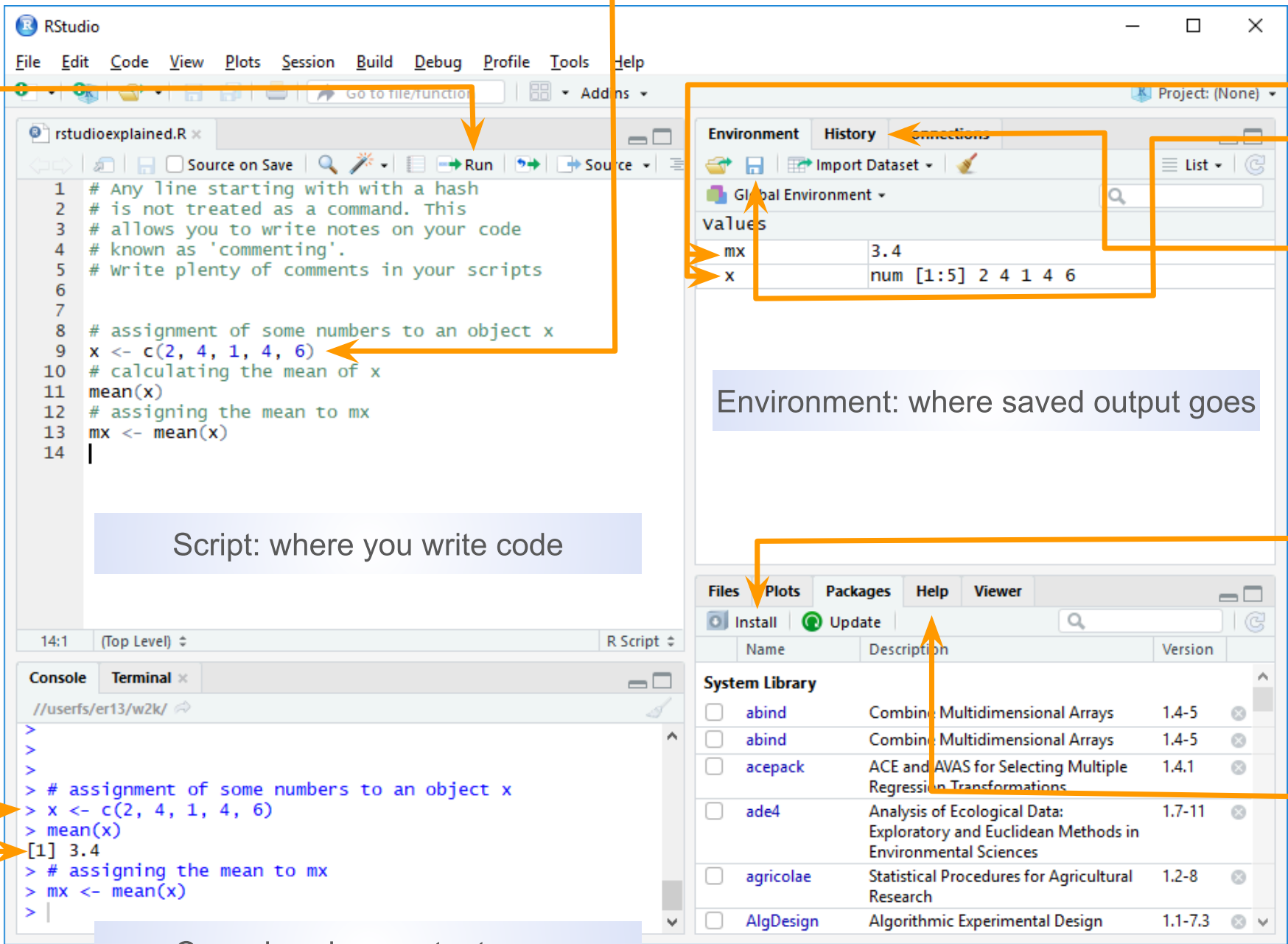
Console: where output goes

Environment

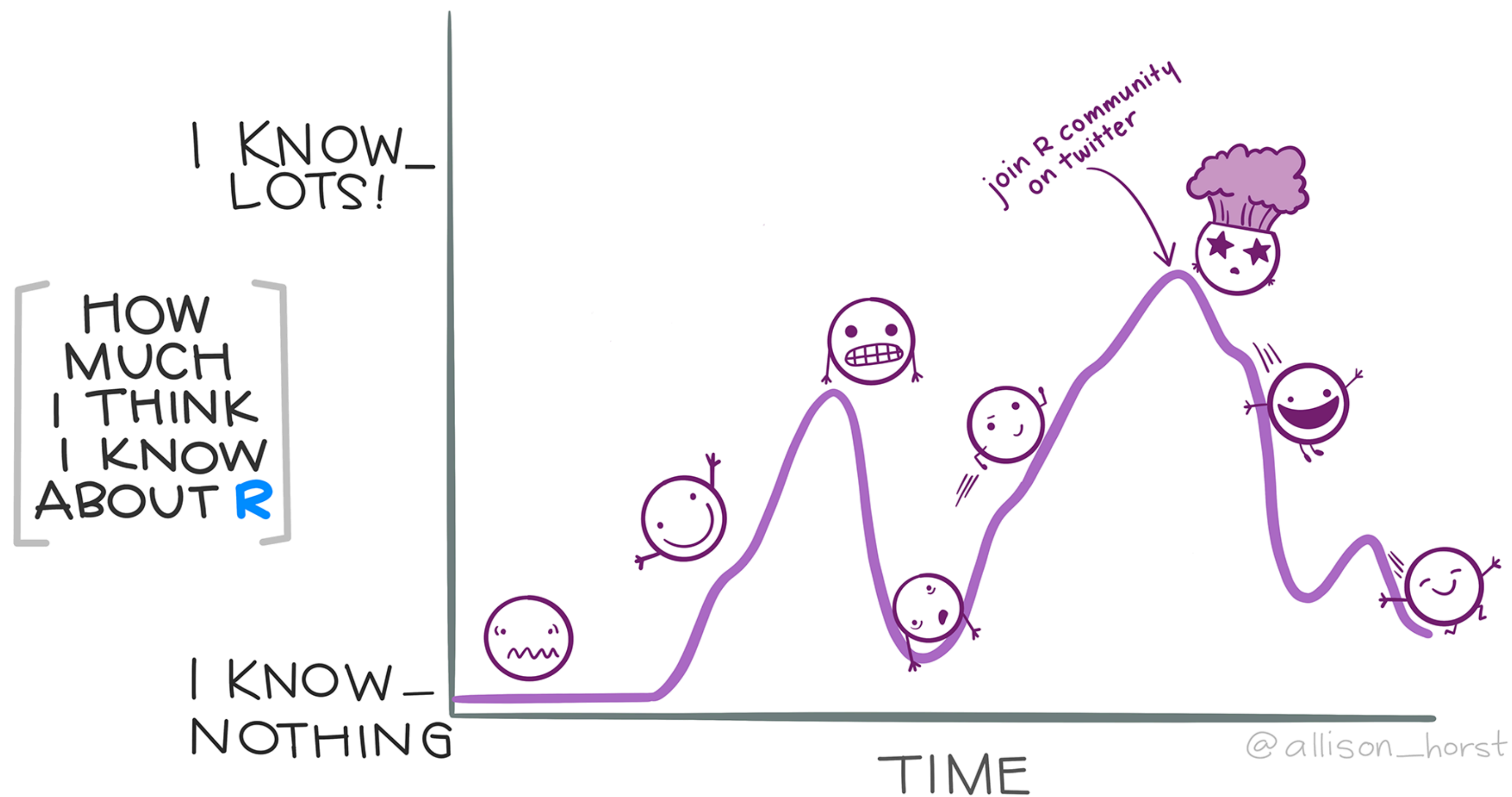
Name objects by assignment to use them again
All the **objects** you created in your session
Saving the environment saves all the objects, but not the code with a **.RData** extension
History
A history of every command you sent to the console, mistakes included.
File can be saved but usually you just need the script

Packages

Many functions come with R
A huge amount of extra functionality is available in packages
Packages can be installed by clicking the Install button
Help
Access to manual pages for all installed packages
Plots
Figure output appears here



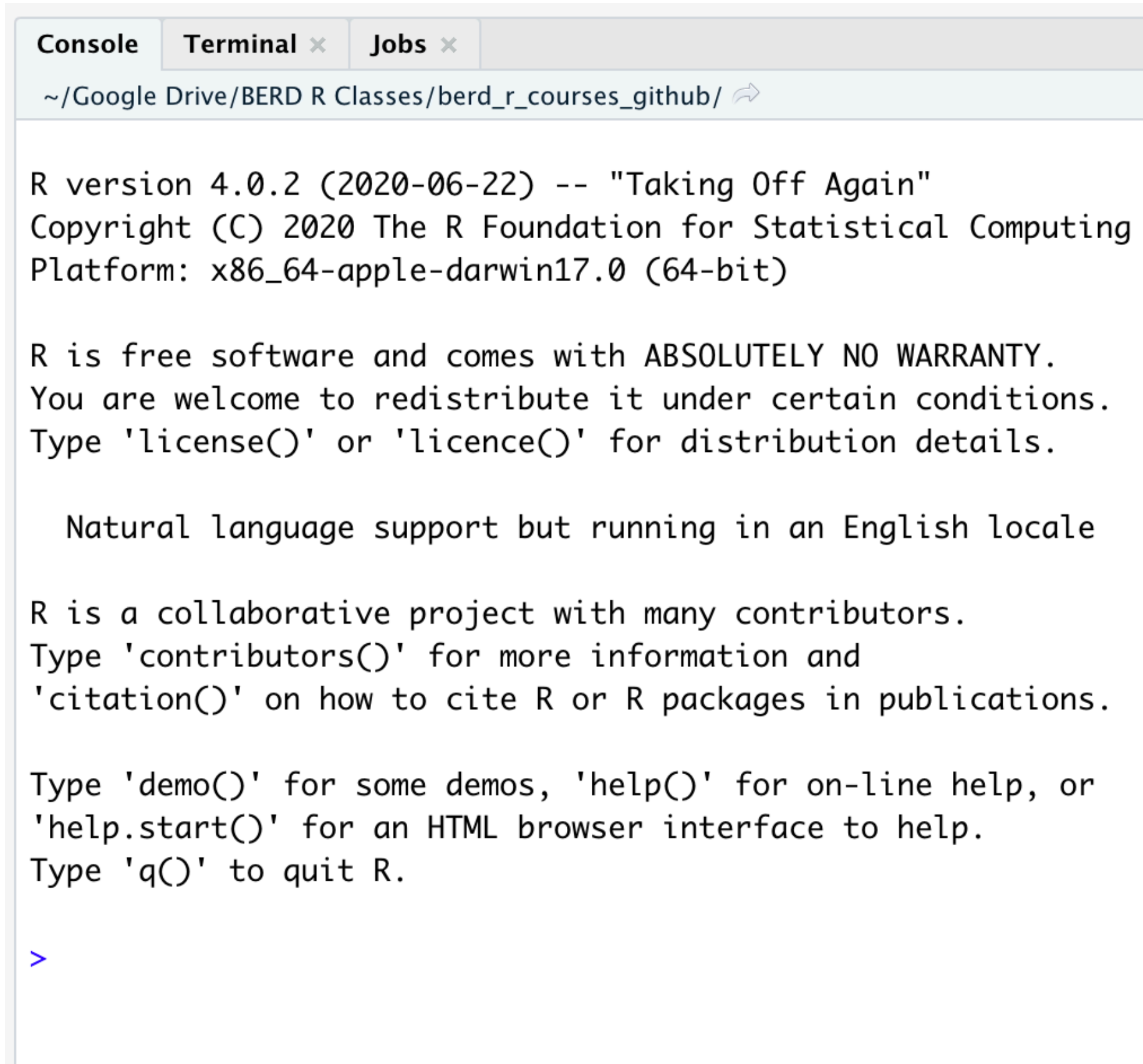
Let's code! R Basics



Artwork by @allison_horst

Coding in the console

When you first open R, the console should be empty.



```
Console Terminal x Jobs x
~/Google Drive/BERD R Classes/berd_r_courses_github/

R version 4.0.2 (2020-06-22) -- "Taking Off Again"
Copyright (C) 2020 The R Foundation for Statistical Computing
Platform: x86_64-apple-darwin17.0 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

>
```

Typing and executing code in the console

- Type code in the console (blue text)
- Press **return** to execute the code
- Output shown below in black

```
> 7
[1] 7
> 3 + 5
[1] 8
> "hello"
[1] "hello"
> # this is a comment, nothing happens
> # 5 - 8
> # separate multiple commands with ;
> 3 + 5; 4 + 8
[1] 8
[1] 12
> |
```

We have an option of where to write our code

- We can use the console directly
 - BUT our work won't be saved
- We can also open up a file called a `.R` file
 - Hit the new document looking button, then click `R script`
 - Now we can type in the R script
 - In the R script, we need to press **cmd + return** or **ctrl + return** to execute the code
 - Output will show up in the Console!
- Example by Nicky here!

Math calculations using R

- We can use R as a calculator!!
- Rules for order of operations are followed
- Spaces between numbers and characters are ignored

```
1 10^2
```

```
[1] 100
```

```
1 3 ^ 7
```

```
[1] 2187
```

```
1 6/9
```

```
[1] 0.6666667
```

```
1 9-43
```

```
[1] -34
```

```
1 4^3-2* 7+9 /2
```

```
[1] 54.5
```

The equation above is computed as

$$4^3 - (2 \cdot 7) + \frac{9}{2}$$